

Lea Smit

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Date of Birth: 14th February 2001
Nationality: South African
Language Skills: Fluent English, Fluent Afrikaans



Professional profile

I am a passionate marine mammal scientist and PhD candidate with hands-on experience in field research, data analysis, and conservation initiatives. Currently conducting research at the University of Pretoria's Mammal Research Institute Whale Unit, I specialise in humpback whale population structure and behaviour. My expertise includes advanced molecular laboratory techniques, CATS and satellite tagging, and bioinformatics pipelines. I am a dedicated researcher, bringing a strong blend of scientific expertise, creativity, and commitment to environmental stewardship, and committed to bridging the gap between genetic data and marine conservation policy.

Key skills

- **Fieldwork & data collection:** Experienced in biopsy sampling, processing, and managing marine mammal field data, photo-identification data, and collection of prey and faecal samples.
- **Statistical & analytical tools:** Proficient in R for statistical analyses, MATLAB for data processing, working knowledge of Bash and Perl; experienced with QGIS mapping, Cervus, and CLC software.
- **Laboratory techniques:** Skilled in molecular methods, including DNA extraction, polymerase chain reaction (PCR), gel electrophoresis, microsatellite genotyping, and mitochondrial DNA sequencing.
- **Communication & outreach:** Strong written and verbal communication skills developed through presenting complex data to international audiences, public engagement, and mentoring undergraduate students.
- **Technical & computer proficiency:** Competent in Microsoft Office Suite (Word, Excel, PowerPoint) for reporting, data management, and presentations.
- **Stakeholder engagement:** Proven ability to collaborate with multidisciplinary teams (DFFE, Whale and Climate Program) and engage in public conservation outreach.

Education and Research

2026 – present **Doctor of Philosophy specialising in Zoology**
Mammal Research Institute Whale Unit, University of Pretoria, South Africa

2024 – 2025 **Master of Science specialising in Zoology (Cum Laude)**
Mammal Research Institute Whale Unit, University of Pretoria, South Africa

2023 **Bachelor of Science Honours – Genetics (Cum Laude)**
University of Pretoria, South Africa

2019 – 2022 **Bachelor of Science – Zoology (extended program)**
University of Pretoria, South Africa

2014 – 2018 **Point High School – Mossel Bay**
Matriculated in 2018 with 4 distinctions

2012 – 2013 **Milkwood Primary School – Mossel Bay**

2006 – 2012 **Al Mawakeb School – Dubai**

Research outputs

- 2024** Presenter at the Whales and Climate Program symposium
- 2025** Oral presentation, 41st Zoological Society of Southern Africa (ZSSA) Conference
- 2025** Oral presentation, Southern African Marine Science Symposium (SAMSS) 2025
- 2026** Poster presentation, Society for Marine Mammalogy (SMM) 26th Biennial Conference on the Biology of Marine Mammals

Teaching and research experience

- 2023-2024** **University tutor – University of Pretoria**
Tutored first- and third-year students in an introductory genetics module and a genome evolution and phylogenetics module.
Key responsibilities:
- Prepared and delivered coursework materials
 - Explained basic scientific concepts
 - Marked semester tests and examinations
- 2025** **Research scientist – (Whale Cruise 2025 – Voyage 306)**
RV: Algoa; Department of Fisheries, Forestry, and the Environment (DFFE)

Academic service & leadership

- 2026 – present** **Local Organising Committee member, Zoological Society of Southern Africa (ZSSA)**
- Co-organising the biennial ZSSA Conference (Swakopmund, Namibia, July 2027)
- 2024 – present** **Student representative, Zoological Society of Southern Africa (ZSSA)**
- Representing student interests on the society council, fostering engagement and supporting student initiatives

Project work and internships

- Dec 2019 – Jan 2020** **Ocean's Research – Mossel Bay, Intern for Marine Field Research**
Completed the competitive Marine Field Research Internship program with Oceans Research, working alongside students from international universities.
Key activities and contributions:
- Participated in multiple ongoing research projects, including shark behaviour, dolphin monitoring, and estuary ecology studies
 - Collected, processed and managed field data following scientific protocols
 - Assisted with community outreach and conservation-related initiatives

Grants and awards

- 2025** 3rd place for “Best Master’s student presentation” at the 41st Zoological Society of Southern Africa (ZSSA) Conference
- 2025** IMarEST Student Bursary for travel to the Southern African Marine Science Symposium
- 2026** The Society for Marine Mammalogy: Student travel grant to the 26th Biennial Conference on the Biology of Marine Mammals

Accomplishments and certificates

- 2013, 2016, 2017** Mossel Bay Eisteddfod (guitar)
- 2016** SA Taalbond
- 2014 – 2017** First aid level 3
- 2017** Bio-Science Global Olympiad
- 2017** Represented South Africa in the Open World Championships for Yachting in Thessalonica, Greece

2018	Assistant Instructor (South African Sailing)
2019	Advanced Open Water Scuba Diver
2019	Marine Mammal Medic
2020	Dive Against Debris
2024	Pipetting Workshop
2025	Personal Survival Techniques training course (exp. 2030)
2025	South African Maritime Safety Authority (SAMSA) Medical Fitness certificate (exp. 2027)

Special interests and experience

2010 – present Horseback riding (outrides, jumping, horsemanship)

2023 – present Padel tennis/jogging/cycling

References

Els Vermeulen – Senior Lecturer and Research Manager at the Mammal Research Institute Whale Unit – University of Pretoria

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Anri van Wyk – Postdoctoral Fellow – University of Pretoria

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